Pricing Prediction Market Odds: Crypto

OQG Group Project | Repo: https://github.com/xa-v1/pm_crypto_project

Project Overview

Strategy area: Alternative data / prediction markets

Research question: Do Kalshi prediction market probabilities for BTC and ETH 15-minute price direction contain exploitable directional signals, and how do market calibration, intraday seasonality, momentum decay, and cross-asset information jointly determine realized prediction accuracy?

Motivation: Kalshi's 15-minute BTC and ETH up/down contracts provide a high-frequency labeled dataset at 1-minute resolution that enables both signal extraction and market microstructure analysis. Initial exploratory work confirms a statistically significant directional signal at threshold 0.55 (cumulative PnL positive across ~900 trades for both assets), meaningful cross-asset correlation ($r = 0.466$ between BTC and ETH P(UP) at open), and a statistically significant cross-asset predictive relationship (ETH P(UP) predicts BTC outcome, $r = 0.148$, $p < 0.001$). This proposal formalizes that work into a rigorous modeling and backtesting framework.

Data

- **Source:** Kalshi API, "BTC Up or Down in 15 minutes?" and "ETH Up or Down in 15 minutes?"
- **Coverage:** February 16 to March 26, 2026; 33,951 rows (BTC) and 29,136 rows (ETH) at 1-minute resolution
- **Unit of observation:** one row per minute per contract; each 15-minute contract window contains up to 15 rows
- **Columns:** Timestamp, P(UP), P(DOWN), Momentum, Prediction, Actual Outcome, Correct?, Market Ticker, Volume
- **Known issues:** data gap from approximately February 28 to March 10; contracts with fewer than 15 rows flagged with an is_clean indicator; P(UP) + P(DOWN) deviations beyond 0.01 tolerance logged

Research Questions and Hypotheses

- Are high-conviction contracts (P(UP) > 0.70) systematically overconfident, and does calibration error vary by time of day? Hypothesis: ETH shows greater overconfidence in high-conviction bins than BTC, and calibration quality degrades near session boundaries.
- Does prediction accuracy vary by hour of contract open? Hypothesis: contracts opening near session boundaries exhibit lower accuracy due to reduced liquidity and informed participation.
- Does the accuracy-conviction relationship evolve over the contract window, and is there an optimal entry minute per conviction level? Hypothesis: high-conviction contracts stabilize earlier than low-conviction contracts.
- Does BTC P(UP) lead ETH P(UP) at sub-minute lags? Hypothesis: BTC leads ETH, reflecting its larger market cap and liquidity.
- Does a cross-asset XGBoost model outperform a same-asset logistic regression? Hypothesis: nonlinear interactions between conviction, momentum, and cross-asset features improve out-of-sample accuracy.

Feature Engineering

Contract-level features

- **P(UP) at open:** probability at minute 0; primary directional signal
- **Momentum (first 3 min):** change in P(UP) from minute 0 to 3; early sentiment drift
- **Mean P(UP) (first 5 min):** smoothed early-window signal
- **Conviction spread:** |P(UP) - 0.5| * 2, normalized to [0, 1]; measures how one-sided the market is at open
- **Hour of open:** extracted from Timestamp; used for intraday seasonality
- **Paired-asset P(UP):** ETH P(UP) at matched contract open time for BTC models, and vice versa
- **Optimal entry minute:** derived from momentum decay analysis; minute at which accuracy plateaus per conviction bin
- **Calibration bin:** P(UP) discretized into 0.05-wide intervals

Minute-level features

- **P(UP) at minute t:** raw within-window probability snapshot
- **Cross-correlation at lag k:** BTC vs ETH P(UP) at lags -3 to +3 minutes across matched contract windows
- **Accuracy by (minute, conviction bin):** binary correct/incorrect label grouped by opening P(UP) bin; produces the decay heatmap

Methodology

Xavi -- signal development, calibration, intraday seasonality

- Build a heatmap of prediction accuracy by (minute within contract, P(UP) bin at open) to characterize how quickly the signal stabilizes as a function of opening conviction.
- Build a conviction-conditional entry rule: for each P(UP) bin, find the earliest minute at which rolling accuracy exceeds 60% for a data-driven delay rule for signal entry.
- Test intraday seasonality by calculating mean accuracy and calibration error grouped by hour of open; accuracy bar chart by hour and flag systematically overconfident hours.
- **Deliverables:** momentum decay heatmap, conviction-conditional entry minute table, intraday accuracy bar chart

Matthew -- market disagreement signal and logistic regression baseline

- Market disagreement signal: plot realized accuracy against conviction spread (|P(UP) - 0.5| * 2) to test whether high-conviction contracts are systematically over- or under-confident
- Fit a logistic regression predicting binary outcome (UP = 1) using same-asset features only: P(UP) at open, momentum in first 3 minutes, hour of open; report coefficients, p-values, accuracy, AUC, and confusion matrix.
- **Deliverables:** disagreement signal plot and logistic regression results table.

Zoe -- lead-lag analysis and XGBoost

- Lead-lag: compute cross-correlation between BTC and ETH P(UP) at lags -3 to +3 minutes across matched contract windows; produce a cross-correlogram with 95% confidence bands; if BTC leads ETH, construct a lagged ETH feature for the BTC model.
- XGBoost: train on the full contract-level feature set (P(UP) at open, momentum, hour, conviction spread, paired-asset P(UP), lagged cross-asset feature if significant); use 5-fold time-series cross-validation; report accuracy, AUC against Matthew's logistic baseline.
- SHAP analysis: compute SHAP values and produce a summary bar plot and beeswarm plot tying the full feature set together.
- **Deliverables:** cross-correlogram, XGBoost results vs. logistic baseline, SHAP plots.

Backtesting Framework

- **Baseline:** enter long if P(UP) > threshold, short if P(DOWN) > threshold, flat otherwise; binary PnL (+1 correct, -1 incorrect); threshold = 0.55, zero transaction costs as reference
- **Transaction cost sensitivity:** re-run at 0.1% and 0.5% per trade; report threshold at which net Sharpe turns negative
- **Threshold grid:** sweep 0.50 to 0.70 in steps of 0.05; report hit rate, trade count, cumulative PnL, and annualized Sharpe for BTC and ETH

Evaluation Metrics

- **Calibration:** calibration error per bin, expected calibration error (ECE)
- **Classification:** accuracy, precision, recall, F1, AUC-ROC for both models; McNemar test for significance of accuracy difference
- **Backtest:** annualized Sharpe, max drawdown, hit rate, net PnL after transaction costs
- **Feature importance:** SHAP summary plot and mean absolute SHAP value per feature
- **Lead-lag:** cross-correlogram with 95% confidence bands, peak lag and p-value

Division of Labor

- **Xavi:** calibration extension by time of day, momentum decay heatmap, conviction-conditional entry rule, intraday seasonality analysis
- **Matthew:** market disagreement signal and entry rule, logistic regression baseline
- **Zoe:** BTC/ETH lead-lag analysis, XGBoost model, SHAP analysis, conviction-conditional and XGBoost-driven backtests, overall backtest framework